

Summary

- Final-year ECE student building systems that bridge hardware and software, custom PCBs, bare-metal firmware, cloud platforms at scale, and real-world IoT deployments with active research in renewable energy.

Education

- **Dr. B. C. Roy Engineering College** Durgapur, West Bengal
B.Tech in Electronics & Communication Engineering (ECE); Final-year (Batch 2026) *Expected Jun. 2026*

Achievements

- **PGET-2026 – Rank 35 in ECE** Jun. 2026
Post-Graduate Entrance Test, MAKAUT (West Bengal) – General Category

Technical Skills

- **Embedded Firmware:** C/C++, Arduino Framework, ESP-IDF, Python (Scripting).
- **Hardware Design:** PCB Layout & Schematic (KiCad, Altium, EasyEDA), Circuit Analysis, Soldering & Rework.
- **Telecom & Networking:** 5G and FTTx deployments, failsafe design, telecom-grade routing & switching.
- **IoT & Cloud:** MQTT, HTTP/REST, Firebase/GCP, AWS, Azure, Docker.
- **Web Development:** JavaScript/TypeScript, Next.js, Express, React, PostgreSQL.
- **Lab Instruments:** Oscilloscope, Multimeter, Logic Analyzer, Bench Power Supply.

Experience

- **BSNL** Burdwan, West Bengal
Telecommunications Intern *Summer 2025*
 - **Network Analysis:** Studied live telecom-grade routing/switching protocols; assisted field engineers in troubleshooting fiber optic signal integrity.
- **Tata Power SDI (PMSGY)** Maithon, Jharkhand
Supervising Engineer Trainee *Winter 2026*
 - **Solar Commissioning:** Supervised solar PV installation and safety compliance; conducted site surveys, panel mounting, and on-site inspections.

Projects

- **800W Synchronous MPPT Controller (ESP32)** Academic
ESP32 • MPPT • Incremental Conductance *Winter 2026*
 - **Overview:** Designed and built an 800W synchronous MPPT solar charge controller targeting rural electrification; achieved high conversion efficiency via INC algorithm.
- **Intelligent Smart Grid & Energy Telemetry Platform** Academic
Edge Computing • Power Metrology • Cloud DB • Web *Fall 2025*
 - **Overview:** MCU-driven prepaid energy meter with real-time cloud telemetry, responsive web dashboard, and remote relay cutoff (< 200ms response).

Research & Publications

- **Weather-Integrated PEMS for Residential Hybrid Solar** Electrical
Under Review – electr-2025-0424 *Winter 2025*
 - **Overview:** Predictive energy management using weather data and smart metering to optimize scheduling in residential hybrid solar systems.
- **ESP32-Based Synchronous MPPT SCC with INC Algorithm** ICICET 2026
Accepted – Speaker Category, Track 7 *Aug. 2026*
 - **Overview:** Presentation at Bolpur, Shantiniketan; demonstrated high conversion efficiency for rural electrification via synchronous MPPT with INC.

Interests

- **Audio stories; DIY electronics projects; Solar & renewable tech; Reading research papers; Open-source; Fixing family electronics; Working on:** <https://toolhub.live>